

SARDAR MD. SAFFAT ZABIN

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SUMMARY

Backend Software Engineer building scalable services with **Spring Boot**, **FastAPI**, and **PostgreSQL**, with experience in **LLM systems**, **RAG**, and **AI agents**. Experienced in deploying production systems on **AWS** using **Docker** and **CI/CD**.

WORK EXPERIENCE

OPlusDojo (Remote)

Toronto, Canada

Software Engineer

Sep 2025 – Feb 2026

- Built an asynchronous document ingestion and **RAG** pipeline using **FastAPI**, **Celery**, **Redis**, **S3**, and **PostgreSQL/pgvector** to process uploaded files, extract structured content and embeddings via **Gemini**, and enable semantic search with LLM-evaluated answer generation.
- Built and optimized backend APIs using **RDS PostgreSQL + pgvector** and **S3 presigned URLs**; reduced **p95 latency from 1851ms → 490ms (73%)** through endpoint restructuring; implemented JWT based authentication using **AWS Cognito**.
- Designed and maintained **development CI/CD pipelines** using GitHub Actions for Docker builds, image tagging, and automated deployments to AWS dev environments.
- Worked on deployment workflows for Dockerized **FastAPI** services in AWS environments; configured two **Application Load Balancers (ALB)** (one for API service, one for MCP service) routing behind **CloudFront** for production traffic handling.
- Built internal dashboard views and reusable components in **React + TypeScript** for platform workflows (ralfh-dev.com).

Apurba Technologies Ltd.

Dhaka, Bangladesh

Full Stack Developer (L-1)

Mar 2025 – Oct 2025

- Built a **JavaFX** desktop tool to convert legacy ASCII Bangla fonts to Unicode with concurrent processing (ExecutorService), progress tracking, and cancellation.
- Extended the tool into a full-stack web platform using **Spring Boot** (backend) and **Angular** (frontend); deployed via **Docker Compose** on a government server (ocr.bangla.gov.bd).
- Designed a configurable cron-based cleanup scheduler in **Spring Boot**, driven by YAML configuration to invalidate and delete temporary OCR files with environment-specific policies (e.g., hourly in dev, weekly at 3AM in prod), eliminating manual storage maintenance.

PROJECTS

UrbanNest: Real Estate Platform | [Live](#) | [Backend](#) | [Frontend](#)

- Designed a layered **Spring Boot** backend with DTO separation, **MapStruct** mapping, **OpenAPI** documentation, and **Flyway**-managed schema evolution on **PostgreSQL (AWS RDS)**.
- Implemented dynamic search using **JPA Specifications** and optimized page queries with **@EntityGraph**, reducing N+1 queries from 81+ per page to 2–3 (**97.5% improvement**).
- Built secure authentication using **Firebase JWT + Spring Security**; protected sensitive data with **AES-256-GCM encryption** and **SHA-512 hashing**; implemented file uploads via **AWS S3 presigned URLs**.
- Integrated **Google Maps Platform** (Maps JS, Places API, Geocoding API) for interactive location selection, reverse geocoding, and nearby place search with client-side caching and radius overlays in the UI (**React + TypeScript**).
- Deployed containerized services using **Docker** via **GitHub Actions** → **ECR** → **EC2**, with **Nginx reverse proxy** and **Let's Encrypt SSL termination**; frontend delivered through **S3 + CloudFront (ACM)** CDN with monitoring via **Prometheus + Grafana**.

SKILLS

Programming Languages: Java, C/C++, Python, TypeScript, SQL

Web & Application Frameworks: Spring Boot, FastAPI, Celery, React, Node.js, Express.js, JavaFX

Databases: PostgreSQL (with pgvector), OracleDB, MySQL

Cloud & Infrastructure: AWS (Cognito, EC2, RDS, S3, CloudFront, ECR, ACM), Google Maps Platform (Maps JS, Places API, Geocoding API), Redis, Nginx, Firebase

DevOps & Observability: Docker, Docker Compose, GitHub Actions, Prometheus, Grafana, Micrometer

AI/LLM: LangChain, LangGraph, Gemini API

Version Control: Git, GitHub

EDUCATION

Bangladesh University of Engineering & Technology (BUET)

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

2020 – 2025